Cone conjecture and Constructibility

Daniil Serebrennikov

Johns Hopkins University, Department of Mathematics

Kawamata–Morrison–Totaro cone conjecture

Let X be a projective Calabi–Yau manifold. Then there exist rational polyhedral cones $\Pi, \Pi' \subset \mathrm{NS}_{\mathbb{R}}(X)$ satisfying the following.

$$\overline{A(X)}^e = \bigcup_{\varphi \in \mathrm{Aut}(X)} \varphi^*\Pi, \quad \mathrm{int}(\varphi^*\Pi) \cap \mathrm{int}(\Pi) = \emptyset \text{ if } \varphi^* \neq id.$$

$$\overline{M(X)}^e = \bigcup_{\varphi \in \mathrm{Bir}(X)} \varphi^*\Pi', \quad \mathrm{int}(\varphi^*\Pi') \cap \mathrm{int}(\Pi') = \emptyset \text{ if } \varphi^* \neq id.$$

Example 1

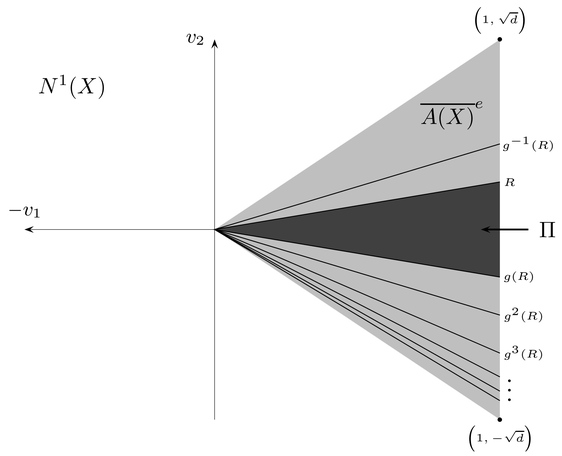


Figure 1:Nef cone of an abelian surface with real multiplication [3].

Theorem A

Let X be a projective Calabi–Yau manifold. Then for every positive integer N the number of orbits for the natural action of $\mathrm{Aut}(X)$ on the set

$$\{A \in \mathrm{NS}(X) : A^{\dim X} = N, \ A \text{ is ample}\}$$

is finite (whenever the defined set is non-empty).

Acknowledgments

The author is grateful to his advisor, Professor Shokurov, for his support and encouragement. This project benefited from thoughtful comments of Philip Engel, Stefano Filipazzi and Nazim Khelifa. The author also thanks the Simons Foundation for this opportunity to present his research.

Kawamata–Matsuki conjecture

The number of isomorphism classes of projective minimal models in a fixed birational class is finite.

Bounded polarization

A collection $\mathfrak{C}$ of projective manifolds X_α of dimension d *admits a bounded polarization* if there exists a positive integer N and very ample line bundles A_α on X_α such that $A_\alpha^d \leq N$ for all X_α in $\mathfrak{C}$.

Example 2

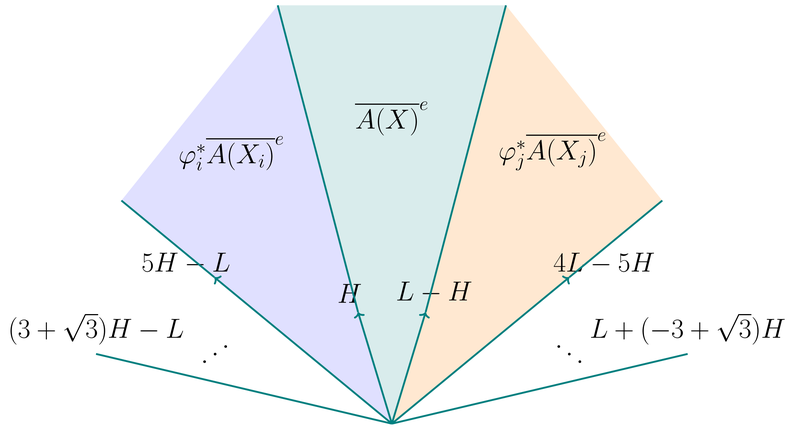


Figure 2:Movable cone of a certain Calabi–Yau 3-fold [2].

$$\overline{M(X)}^e = \bigcup_{\varphi_i \colon X \dashrightarrow X_i} \varphi_i^*\overline{A(X_i)}^e$$

Theorem B

Let $\mathfrak{C}$ be a collection of projective birationally equivalent Calabi–Yau manifolds. Suppose $\mathfrak{C}$ admits a bounded polarization. Then the number of isomorphism classes in $\mathfrak{C}$ is finite.

References

[1] L. D'Urso, *On the nef cones of blowups of the projective plane*. ArXiv e-print.

[2] C.-J. Lai and S.-S. Wang, *The movable cone of certain calabi–yau threefolds of picard number two*, J. Pure Appl. Algebra **226** (2022), no. 2, 106841.

[3] A. Prendergast-Smith, *The cone conjecture for abelian varieties*, J. Math. Sci. Univ. Tokyo **19** (2012), no. 2, 243–261.

[4] B. Totaro, *The cone conjecture for calabi–yau pairs in dimension 2*, Duke Math. J. **154** (2010), no. 2, 241–263.

Conjecture

Let X be a projective minimal model, and $f : \mathcal{X} \to S$ be a projective family of minimal models. Then the subsets

$$\mathrm{Iso}_X(\mathcal{X}/S) = \{s \in S : \mathcal{X}_s \approx X\},$$
$$\mathrm{Bir}_X(\mathcal{X}/S) = \{s \in S : \mathcal{X}_s \sim_{\mathrm{bir}} X\}$$

are Zariski constructible in S.

Non-example 1

Let S be a smooth projective K3 surface such that $\mathrm{Aut}(S)$ is infinite. Consider the trivial family $p_2 : S \times_{\mathbb{C}} S \to S$. Let $\mathcal{X} = \mathrm{Bl}_\Delta(S \times_{\mathbb{C}} S)$ be the blow-up of the diagonal $\Delta \subset S \times_{\mathbb{C}} S$, and put $X = \mathcal{X}_{s_0}$ for some very general point $s_0 \in S$. Then $\mathrm{Iso}_X(\mathcal{X}/S)$ is an infinite countable set that is not constructible.

Non-example 2

Let $C \subset \mathbb{P}^2_{\mathbb{C}}$ be a smooth cubic passing through 9 very general points $p_i \in \mathbb{P}^2_{\mathbb{C}}$. Let B be the strict transform of C to $X = \mathrm{Bl}_{p_1,p_2,\ldots p_9}\mathbb{P}^2_{\mathbb{C}}$. Then (X, B) is a plt Calabi–Yau surface, and $\mathrm{Aut}(X) = \{\mathrm{id}\}$. Nevertheless, $\overline{A(X)}$ admits a rational polyhedral fundamental domain for the Cremona action [1]. Then there are infinitely many very ample line bundles A_i on X such that $A_i^2 = N$ for some N. Using the Hilbert–Chow schemes argument one can construct a family $(\mathcal{X}/S, \mathcal{B})$ such that $\mathrm{Iso}_{(X,B)}(\mathcal{X}/S, \mathcal{B})$ is not constructible.

Theorem C

Let X be a Calabi–Yau manifold, and $f : \mathcal{X} \to S$ be a projective family of Calabi–Yau manifolds. Then the subsets

$$\mathrm{Iso}_X(\mathcal{X}/S) = \{s \in S : \mathcal{X}_s \approx X\},$$
$$\mathrm{Bir}_X(\mathcal{X}/S) = \{s \in S : \mathcal{X}_s \sim_{\mathrm{bir}} X\}$$

are Zariski constructible in S.

Proof Sketch

We can assume $\mathrm{Bir}_X(\mathcal{X}/S)$ is dense in S. Consider a period map $\Phi^{\mathrm{CY}} : S \to \Gamma\backslash\mathbb{D}$ associated with a polarized $\mathbb{Z}$-VHS V^{tr}, the transcendental part of $R^d f_*\mathbb{Z}$. Since $H^{d,0}(X) = H^0(X, \omega_X) \approx \mathbb{C}$ is a birational invariant, it follows that the transcendental part for $H^d(X, \mathbb{Z})$ is also a birational invariant. So, $\Phi^{\mathrm{CY}}(\mathrm{Bir}(\mathcal{X}/S)) = \{\star\}$. Hence, by the infinitesimal Torelli theorem, $\mathcal{X}/S$ is isotrivial.